

PageRank

Summary Chapter 5.1.

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Agenda

Chapter 5.1.

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Introduction



Search Engines

Basic problem:



A library with billions
of documents



No centralized
organization



No
librarian



Anyone can add
documents

- Search engines should be responsible for locating the most relevant documents and web pages.
- Usually worked by crawling the Web and listing the found in each page, in an **inverted index**.
- When a search query issued, pages with those terms were extracted from the inverted index.
- Ranked in a way that reflected **the use of the terms within the page**.
- **Problem1:** Presence of a term in a header of the page made the page more relevant.
- **Problem2:** Term spam.

Token	Document ID
Advanced	1, 2, 3
Big	2, 3
Data	2, 3, 4
Analysis	3, 4

An efficient and accurate web search is needed!

5.1.

PageRank

5.1.1. Early Search Engines and Term Spam

- Search engines like Google combat **spammers** who had previously rendered search nearly ineffective.
- **PageRank** was the first algorithm used by Google to order search results.
- It is a **link analysis algorithm** (that makes use of the eigenvalue problem).
- To combat term spam, Google introduced two innovations



- The simulation of **random surfers** helps approximate the "importance" of web pages based on two main ideas:
 - **User Behavior**
 - **Surfing Patterns**



- Counting only the number of incoming links (**Simplified PageRank**), to estimate page importance is ineffective because of **fake link networks**.
- So, the content of a page was judged not only by the terms appearing on that page, but by the **terms used in or near the links to that page**

5.1.2 Definition of PageRank: Idealized PageRank

- The probability distribution for the location of a random surfer can be described by a **column vector** whose j_{th} component is the probability that the surfer is at page j .
- This probability is the **idealized** PageRank function
- $v = \lambda M v$
- $\lambda = 1$
- **Markov Processes**
- The limit is reached when multiplying the distribution by M another time does not change the distribution
 - In practice, for the Web itself, 50–75 iterations are sufficient
- the limiting v is an eigenvector of M

$$v = \begin{pmatrix} p_1 \\ p_2 \\ p_3 \\ \dots \\ p_j \end{pmatrix}$$

column vector example

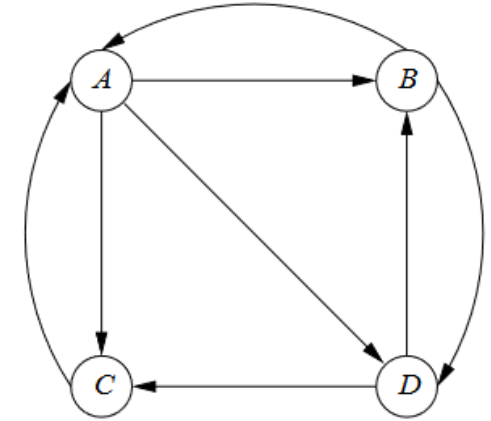


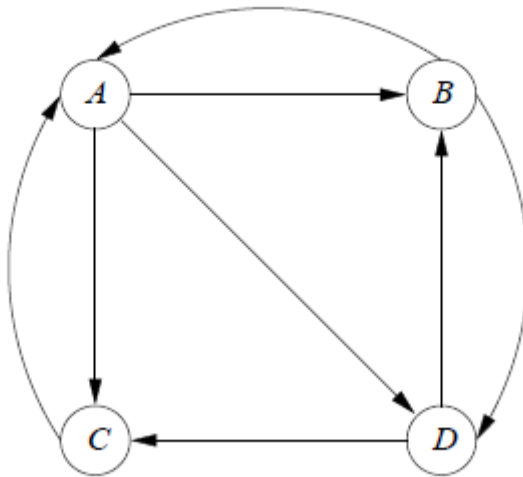
Figure 5.1: A hypothetical example of the Web

$$M = \begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix}$$

Example 5.1: The transition matrix for the Web of Fig. 5.1 is

Digression: Solving Linear Equations

- Looking at this 4-node Graph, you might think that the way to solve the equation $v = Mv$ is by **Gaussian elimination**.
- However, in realistic examples, where there are **tens or hundreds of billions of nodes**, Gaussian elimination is **not feasible** $O(n^3)$.
- Thus, the only way to solve equations on this scale is **to iterate**.
- $v = Mv$ has an infinite number of solutions
- When we include the constraint that **the sum of the components is 1**, as we have done, then we get a unique solution.



$$v = \begin{pmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{pmatrix}$$

$$M = \begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}, \begin{bmatrix} 9/24 \\ 5/24 \\ 5/24 \\ 5/24 \end{bmatrix}, \begin{bmatrix} 15/48 \\ 11/48 \\ 11/48 \\ 11/48 \end{bmatrix}, \begin{bmatrix} 11/32 \\ 7/32 \\ 7/32 \\ 7/32 \end{bmatrix}, \dots, \begin{bmatrix} 3/9 \\ 2/9 \\ 2/9 \\ 2/9 \end{bmatrix}$$

$$v \times M$$

$$v \times M^2$$

$$v \times M^3$$

$$v \times M^4$$

$$v \times M^n$$

5.1.3 Structure of the Web: The *bowtie*

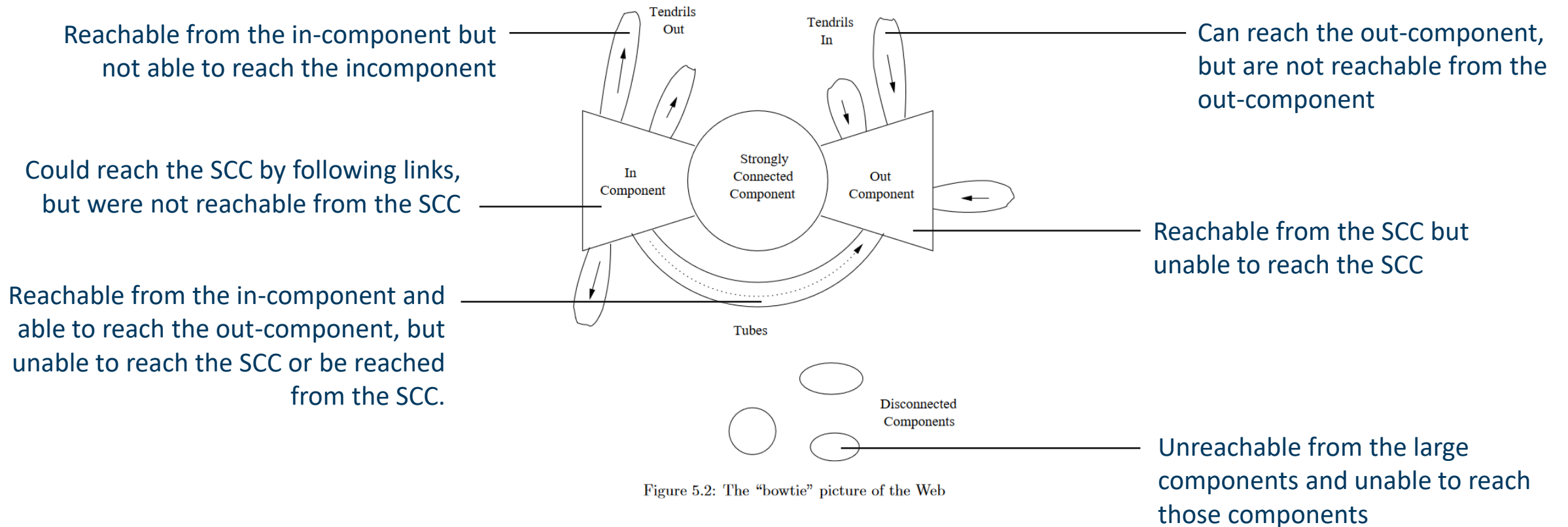


Figure 5.2: The “bowtie” picture of the Web

The bowtie structure violates the Markov property

5.1.3 Structure of the Web

There are really two problems we need to avoid:

Dead End: A page that has no links out.

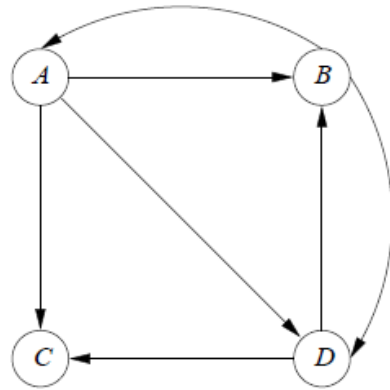


Figure 5.3: *C* is now a dead end

Spider traps: Groups of pages that all have outlinks but they never link to any other pages.

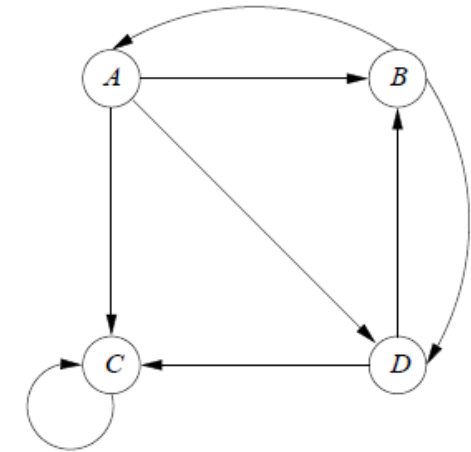


Figure 5.6: A graph with a one-node spider trap

5.1.4 Avoiding Dead Ends

There are two approaches to dealing with dead ends.

1. Dropping the dead ends and their incoming arcs.
2. Modify the process by which random surfers are assumed to move about the Web (**taxation**).

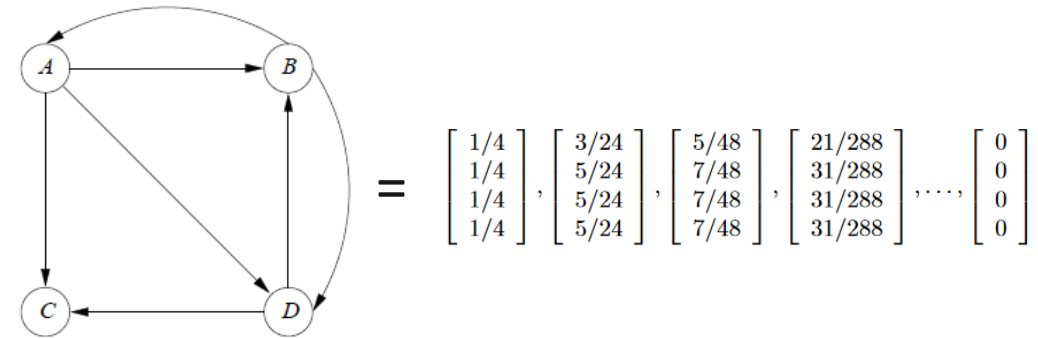


Figure 5.3: *C* is now a dead end

Calculating the PageRank for the dropped ends

- Looking at the predecessors: A and D
- **A** has three successors: contributes $\frac{1}{3}$ of its PageRank ($= \frac{2}{9}$) to **C**
- **D** has two successors: contributes $\frac{1}{2}$ of its PageRank ($= \frac{3}{9}$) to **C**
- Thus, the PageRank of C is $\frac{1}{3} \times \frac{2}{9} + \frac{1}{2} \times \frac{3}{9} = \frac{13}{54}$
- E has only one predecessor, C so it has the same PageRank as C
- **Note that the sums of the PageRanks exceed 1**

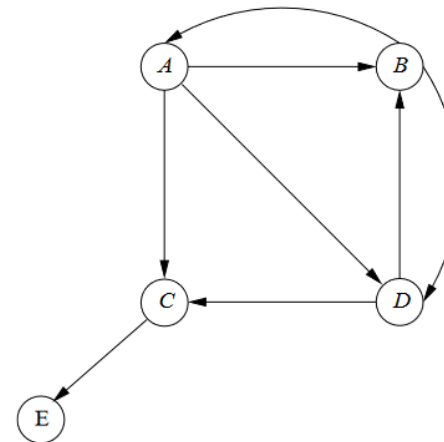


Figure 5.4: A graph with two levels of dead ends

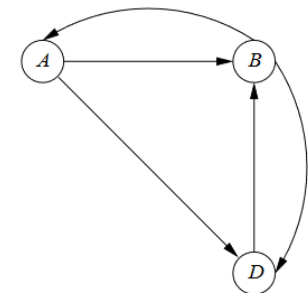


Figure 5.5: The reduced graph with no dead ends

5.1.5 Spider Traps and Taxation

Spider Traps: A set of nodes with no dead ends but no arcs out

Solution: allowing each random surfer a small probability of teleporting to a random page, rather than following an out-link from their current page → **TAXATION**

$$v' = \beta Mv + \frac{(1 - \beta)e}{n}$$

- β is a chosen constant, usually in the range 0.8 to 0.9
- e is a vector of 1's for each node in the graph
- n is the number of nodes in the Web graph
- βMv : Random surfer decides to follow an out-link from their present page
- $(1 - \beta)e/n$: New random surfer at a random page.

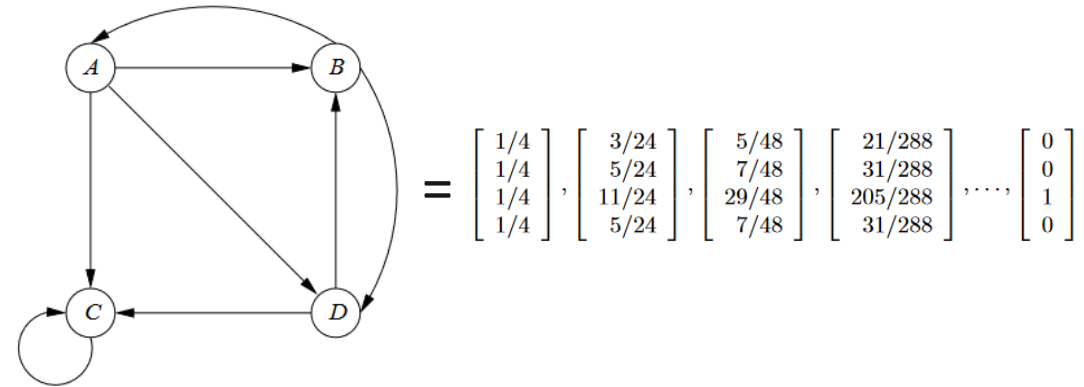


Figure 5.6: A graph with a one-node spider trap

Example 5.6: $\beta = 0.8$

$$v' = \begin{bmatrix} 0 & 2/5 & 0 & 0 \\ 4/15 & 0 & 0 & 2/5 \\ 4/15 & 0 & 4/5 & 2/5 \\ 4/15 & 2/5 & 0 & 0 \end{bmatrix} v + \begin{bmatrix} 1/20 \\ 1/20 \\ 1/20 \\ 1/20 \end{bmatrix}$$

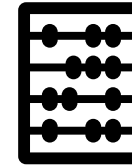
$$= \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}, \begin{bmatrix} 9/60 \\ 13/60 \\ 25/60 \\ 13/60 \end{bmatrix}, \begin{bmatrix} 41/300 \\ 53/300 \\ 153/300 \\ 53/300 \end{bmatrix}, \begin{bmatrix} 543/4500 \\ 707/4500 \\ 2543/4500 \\ 707/4500 \end{bmatrix}, \dots, \begin{bmatrix} 15/148 \\ 19/148 \\ 95/148 \\ 19/148 \end{bmatrix}$$

Conclusion



5.1.6 Using PageRank in a Search Engine

- Google uses **250+ page properties** to decide the order of results.
- Pages **must contain at least one search term to be ranked**.
- Pages missing all search terms are unlikely to appear in top results.
- **A score is computed for each qualified page**, key factors are besides others:



PageRank



Search Terms

Thank You!

Any Questions?